NanoSphere
Nanotechnology SaaS Platform

Project Report | Version 1.0 | 2026-10-11

Learning + AI + Business/Industrial + Simulation & Modeling

Full-stack workspace: React + Vite + TypeScript + Tailwind frontend, FastAPI + SQLAlchemy backend (PostgreSQL prod / SQLite dev), JWT auth, real payment-gated plans, database-backed catalogue, courses, research, tickets and admin.

1. Platform Purpose

NanoSphere serves four pillars: (1) Learning & Education SaaS - 4 courses, 6 lessons, quizzes, email-gated enrollment, limited free offer then paid, glossary in DB. (2) AI Nanotechnology Platform - material recommender, property predictor, literature miner. (3) Business/Industrial SaaS - batches, QC, cost estimator, 8 guided lab project templates with a 6-week roadmap. (4) Simulation & Modeling - transport runs with server quotas plus a unit-cell modeling playground and 6 bench calculators.

2. Technology Stack

Layer	Technology	Notes
Frontend	React 19 + Vite 7 + TS 5.9	Strict mode, alias @/*
Styling	Tailwind CSS v4	Custom theme tokens
3D	react-three-fiber + three	Per-material structures
Backend	FastAPI + Uvicorn	OpenAPI at /docs
ORM / DB	SQLAlchemy 2 + Postgres / SQLite	18 tables, seeded catalogue
Auth	bcrypt + JWT (HS256, 24h)	Guests restricted
Science	NumPy/SciPy/pandas	Demo models, disclaimed

3. Feature Checklist (all verified working)

- [OK] Login/Signup - secure auth + 6-digit email verification (15-min expiry).
- [OK] Nanomaterials Explorer - Postgres search/filter/sort + offline fallback.
- [OK] Compare - side-by-side stored-property table (up to 3 materials).
- [OK] Analytics & Charts - uploads parsed with pandas + stats.
- [OK] Research Reports - CSV export + Print/Save-as-PDF report.
- [OK] Historical Data - timestamped experiment runs in DB + UI history.
- [OK] Research - 6 verified DB articles + live Semantic Scholar lookup.
- [OK] Learning Center - DB lessons, quizzes, progress + 18-term glossary.
- [OK] Pricing - Starter $0 / Pro $199 / Team $999, payment-gated server-side.
- [OK] Contact/Support - validated tickets in DB, tracking + admin triage.
- [OK] Admin Panel - protected stats, users (Pro/verify/admin), tickets.

4. Frontend Routes (21/21 return HTTP 200)

Route	Page
/	Dashboard (hero, quick access, trending 3D, activity
/login, /signup	Glass sign-in (backend auth)
/explore	DB catalogue + compare
/materials/:id	Detail + interactive 3D
/simulation	Quota-gated console (5/day Starter)
/modeling	Unit-cell playground
/lab-tools	6 calculators with SVG schematics
/analysis	Charts, CSV/PDF reports, run history
/projects	CRUD + 8 lab templates (email-gated)
/learn	Courses (enroll/pay) + glossary
/ai	Recommender + predictor + miner
/business	Batches, QC, cost estimator
/resources, /docs	Resources + documentation system
/research	Verified catalogue + live lookup
/community	Forum threads, replies, likes
/pricing	Starter/Pro/Team checkout
/contact	Support tickets
/admin	Protected admin (admins only)
/settings	13 tabs: profile->about
/activity	Full activity feed

5. Backend API (representative)

Method + Path	Auth	Purpose
POST /auth/signup|login|guest	no	Accounts + JWT
POST /auth/request|confirm-code	member	Email verification
GET /materials	any	DB catalogue search
POST /science/simulate	any	Run + quota (5/day free)
GET /science/runs	any	Experiment history
GET/POST /projects...	member-write	Projects + tasks
POST /datasets/upload	any	CSV/JSON + pandas stats
GET /learn/*	member	Courses, enroll, progress, glossary
GET/POST /billing/*	member	Status, $199/$999 checkout
GET/POST /support/tickets	public/mine	Contact tickets
GET/PATCH /admin/*	admin	Stats, users, tickets
GET /research/live	any	Semantic Scholar proxy

6. Database (18 tables, integrity OK, 0 orphans)

users, projects, tasks, datasets, activities, notifications, saved_materials, search_history, enrollments, course_progress, purchases, email_tokens, materials(12), articles(6), lessons(6), glossary(18), sim_runs, tickets. Migrations via PRAGMA ALTER block; seeds idempotent.

7. Security Controls

- [OK] Rotated 64-char JWT secret; HS256 pinned (no 'none' attacks); 24h expiry.
- [OK] bcrypt hashing (72-byte cap handled); email format + password rules validated.
- [OK] Guests blocked (403) from projects writes, enrollments, billing, courses, admin.
- [OK] Owner-scoped queries everywhere; admin self-demote blocked.
- [OK] Upload allowlist (.csv/.tsv/.json/.txt) + 10 MB cap; research proxy URL fixed.
- [OK] CORS allowlist; DEV_MAILBOX=false in production; SMTP hook ready.

8. Ve

Suite	Result
backend/verify_backend.py	ALL OK
backend/verify_learn.py	DONE (401/403 gates hold)
backend/verify_billing.py	Quota 5/day, 402, checkout, cancel OK
backend/verify_tiers.py	Pro $199 / Team $999 OK
backend/verify_full.py (24 checks)	ALL OK
backend/verify_adversarial.py	20/20 malicious inputs rejected
backend/verify_security.py	7/7 (forgery, guests, exe upload)
tsc -b + vite build	0 errors, success
Route smoke (21)	21/21 HTTP 200

9. Deployment

See DEPLOY.md. One-command stack: Postgres 16 + FastAPI + nginx static frontend via docker compose (plus managed-host alternative). Production checklist: fresh JWT_SECRET, real CORS_ORIGINS, DEV_MAILBOX=false, ADMIN_EMAILS, HTTPS, DB backups, SMTP.

Disclaimer: catalogue values, indices and simulation curves are illustrative demo models - never use them for engineering, medical or safety decisions.